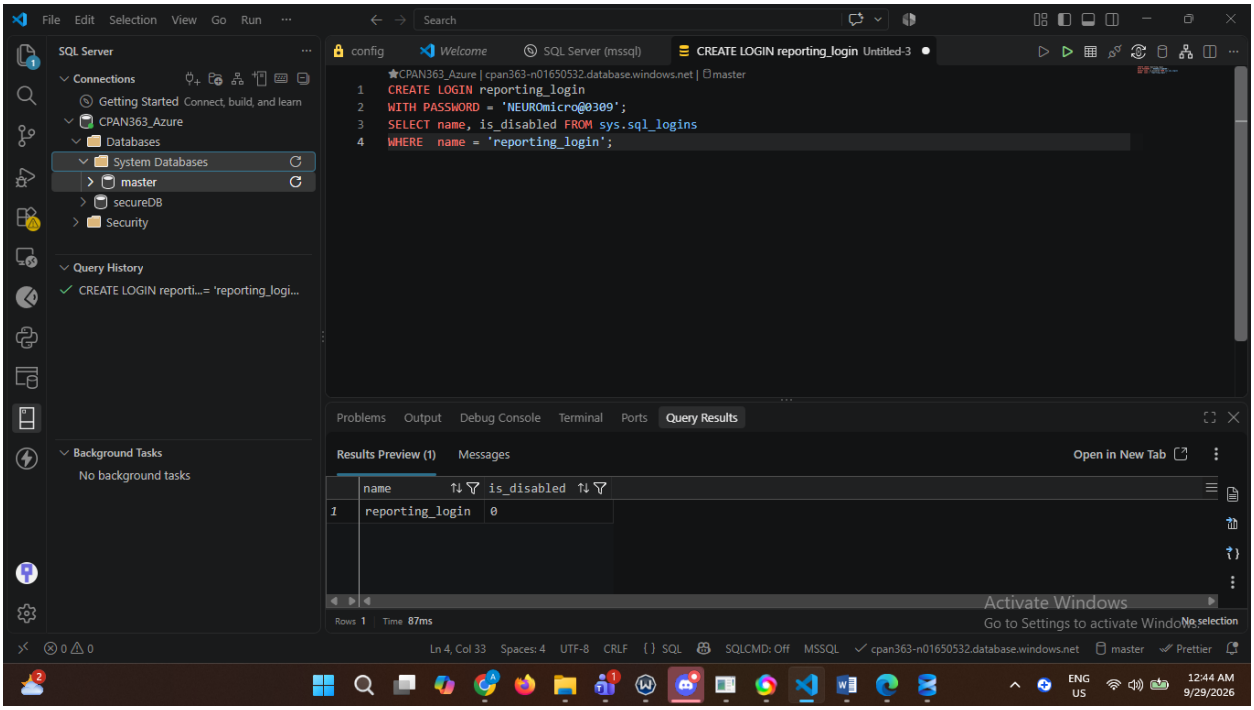
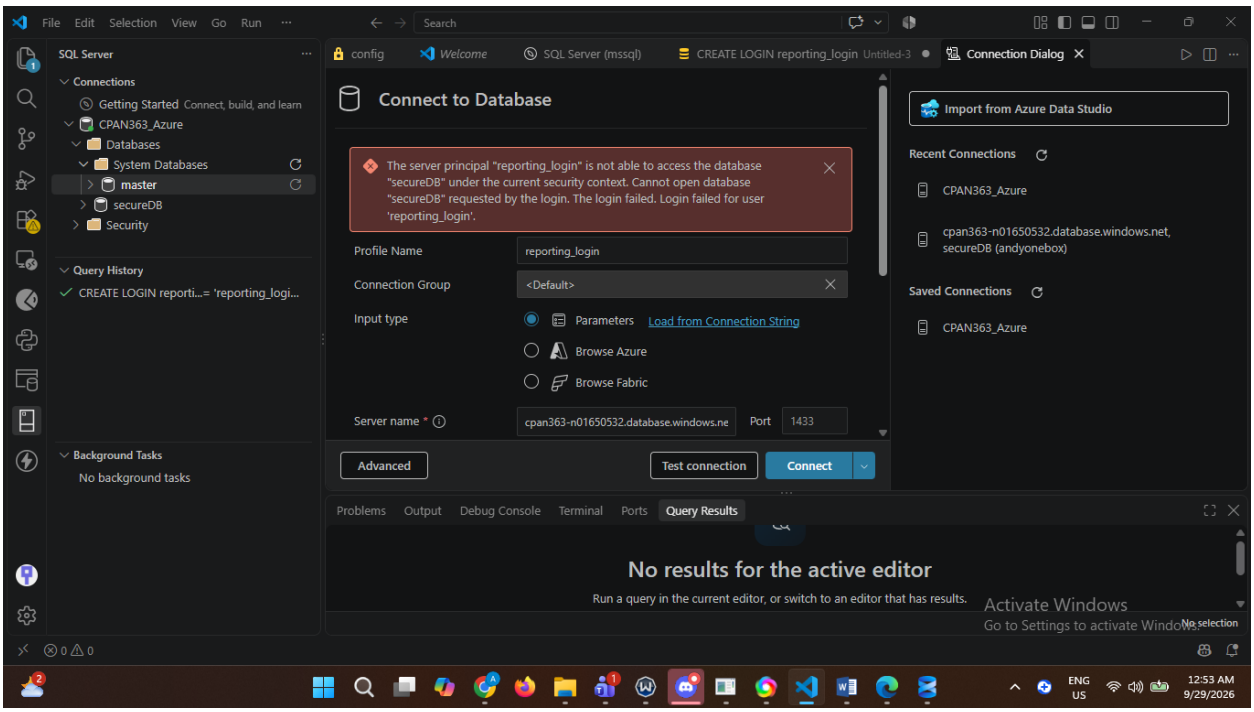
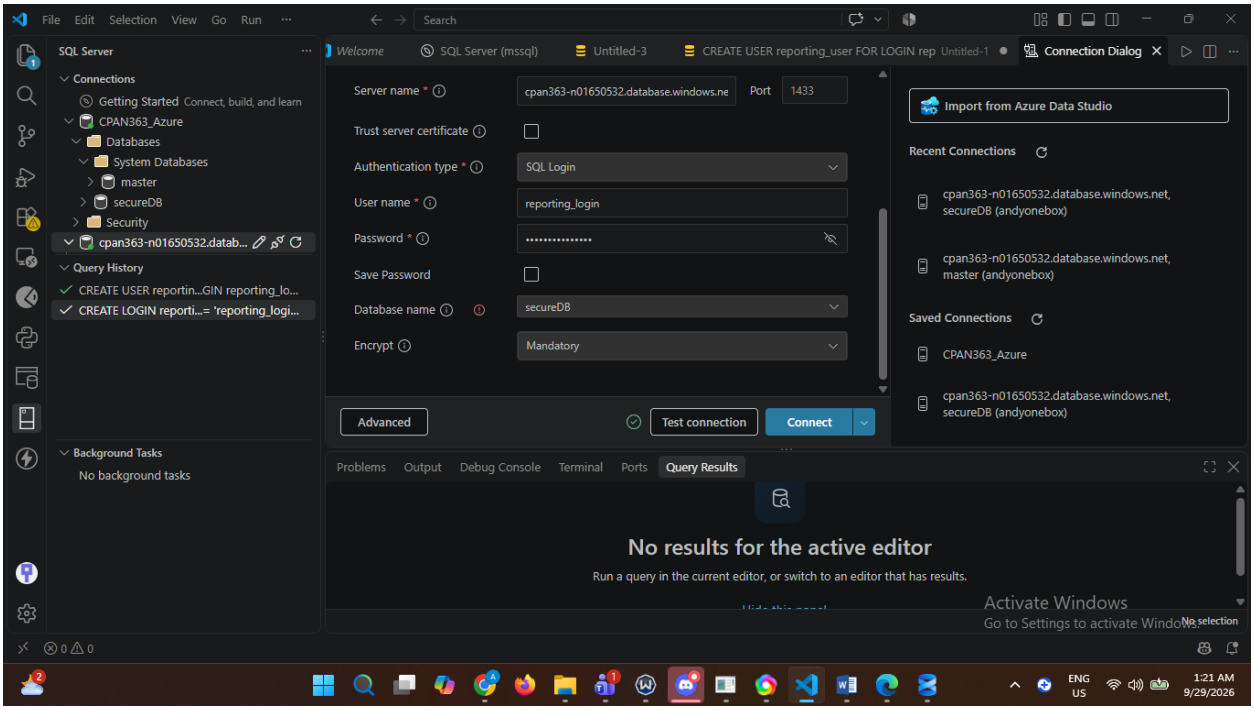
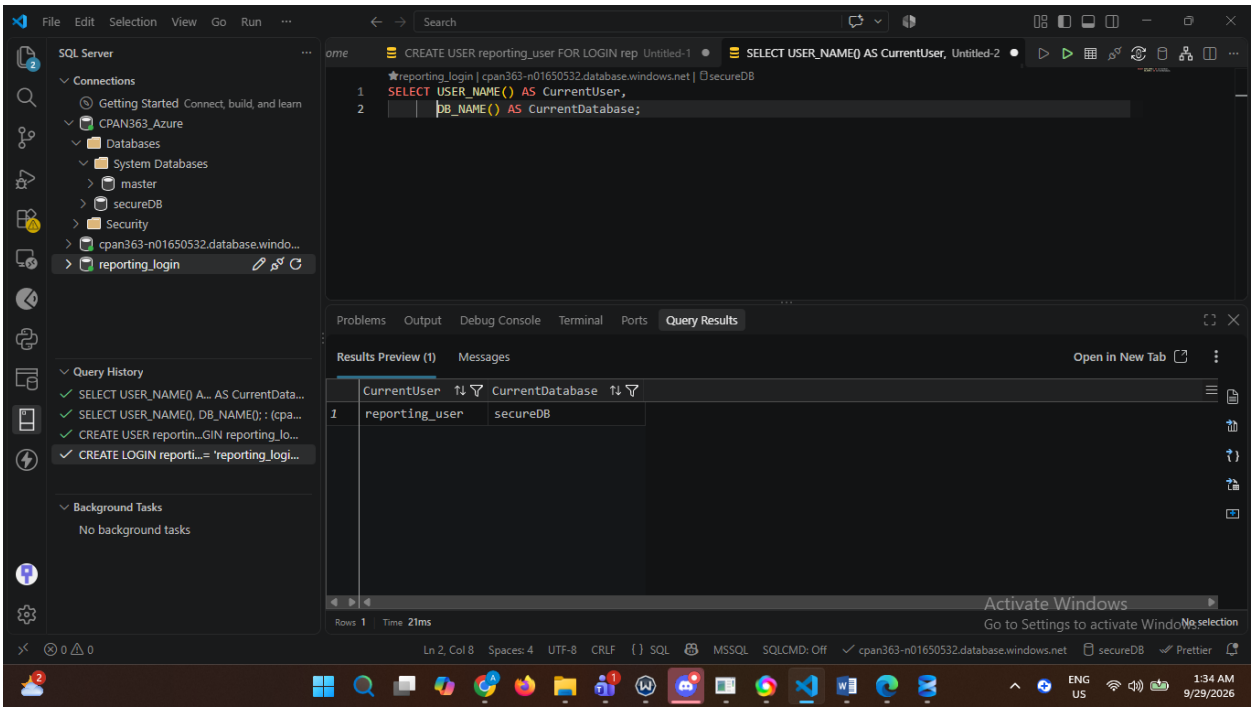
Part 1 — Make the classic mistake, then fix it





Reporting_login test connection succeeded

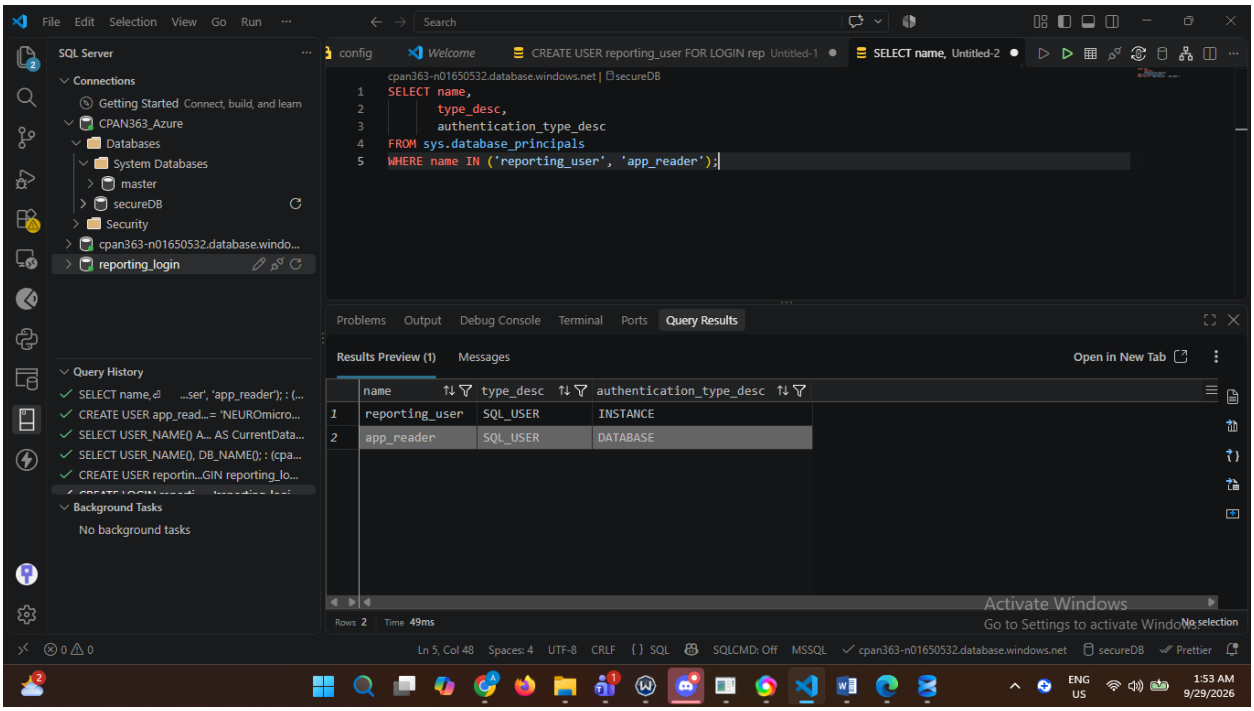




Explanation

The login was created at the server level, so the password was correct and the login could authenticate to the server. However, there was initially no user associated with that login inside the SecureDB database. Creating **reporting_user** linked the login to a database user, allowing access to SecureDB.

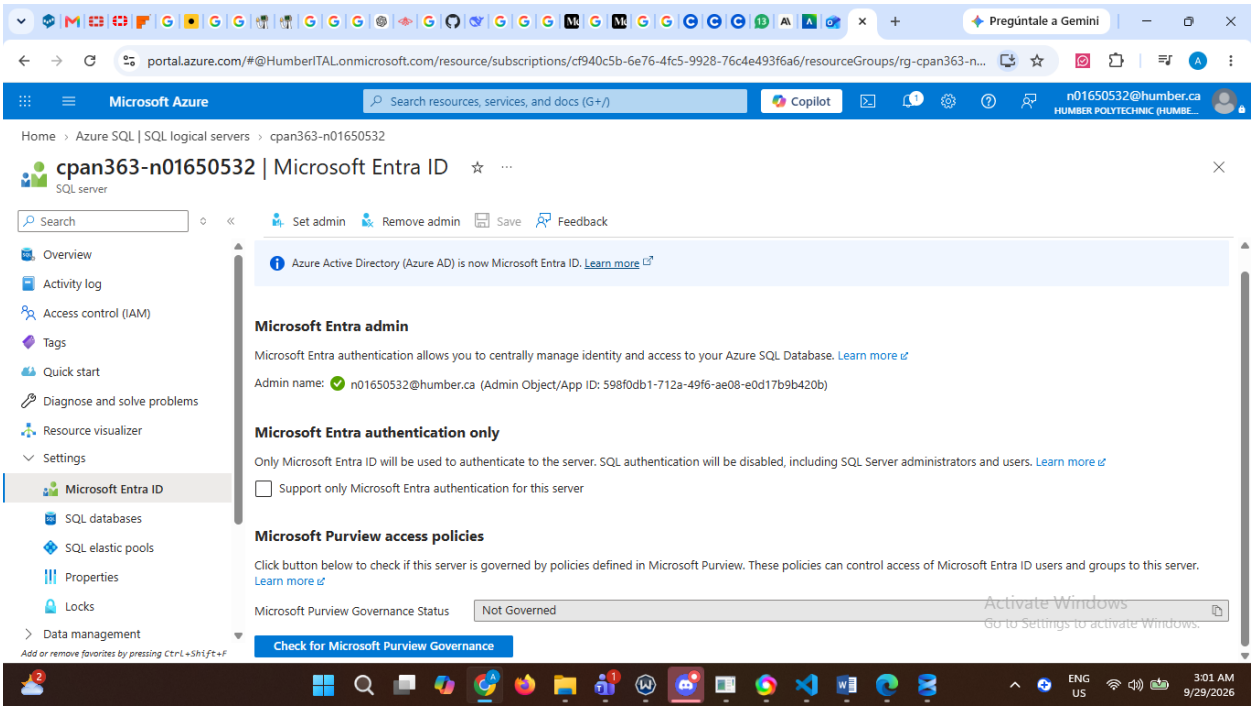
Part 2

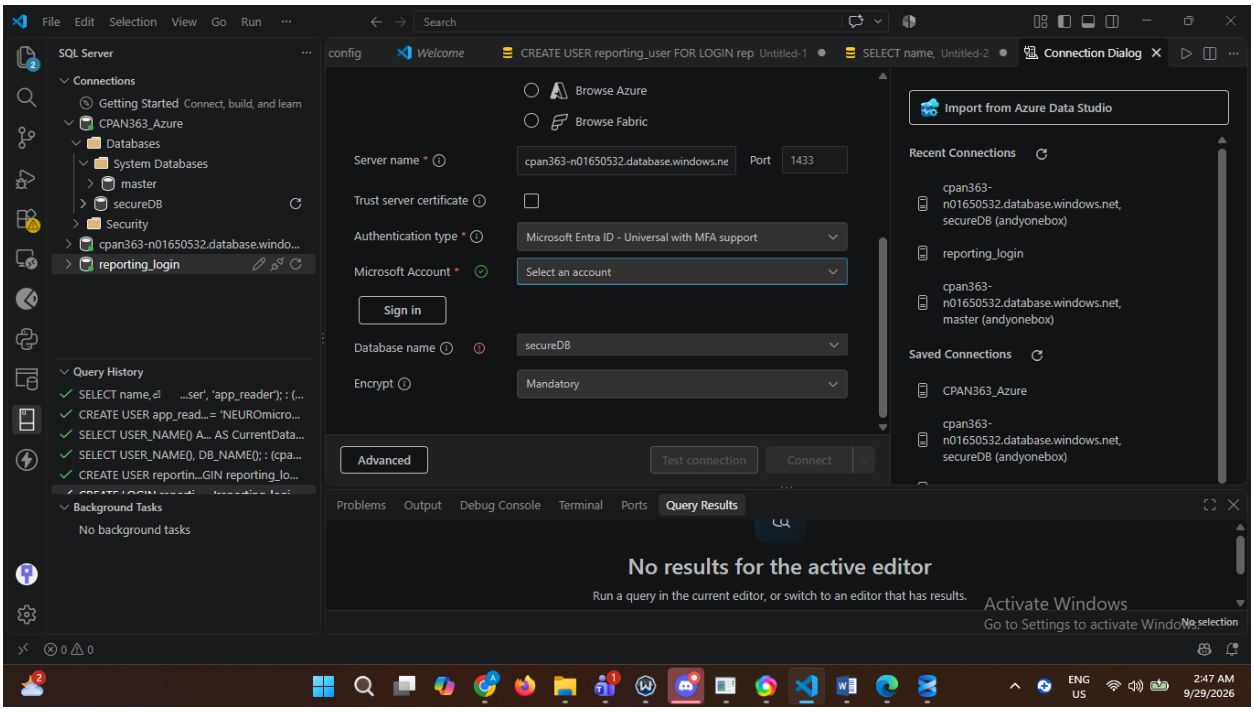


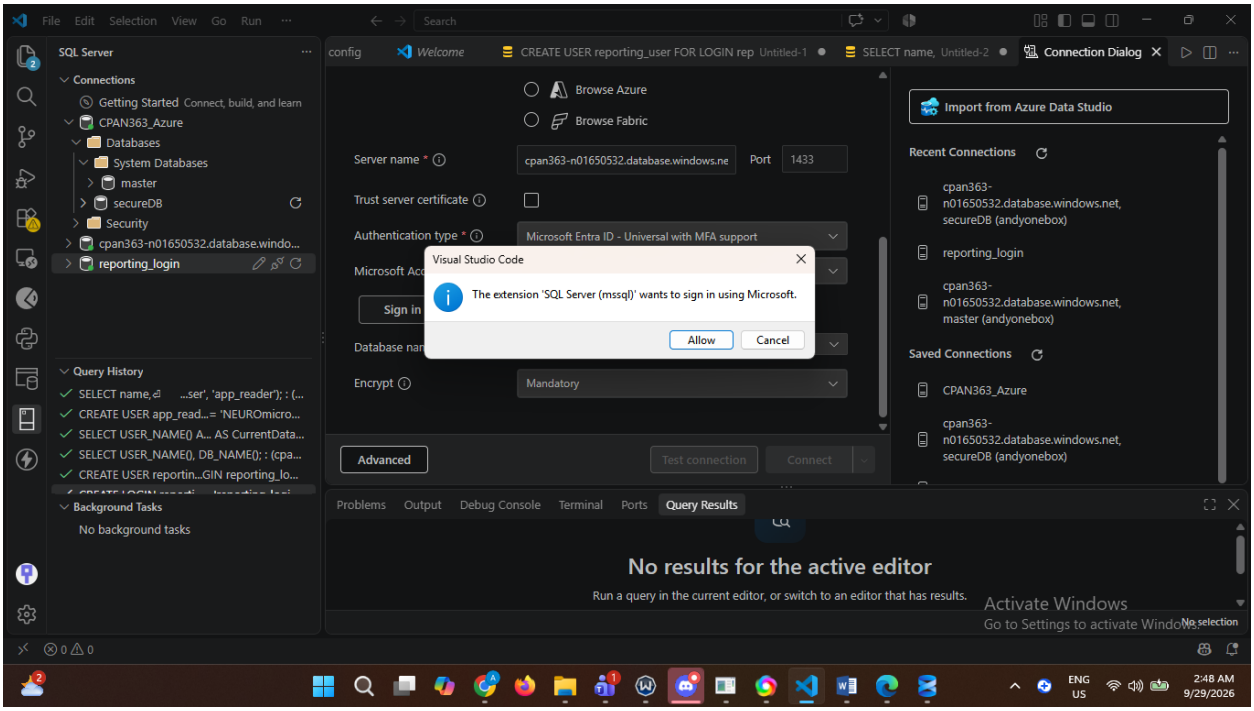
Part 2.3

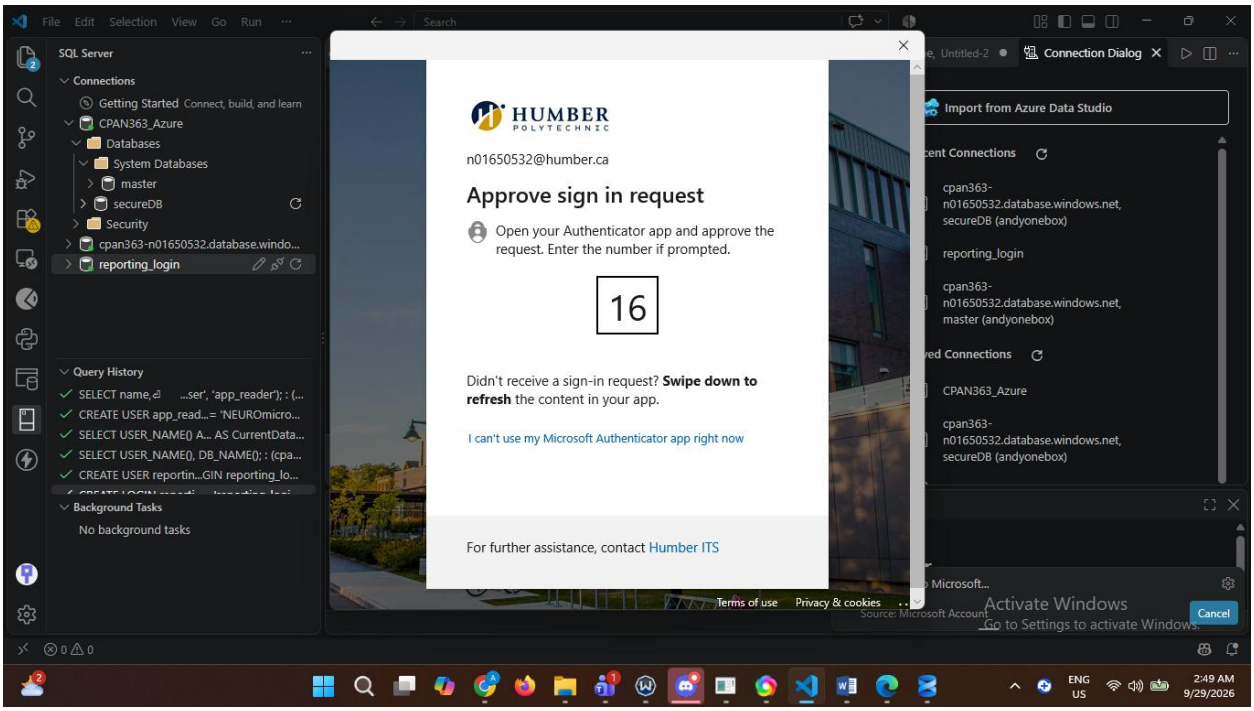
`app_reader` would still work if copied to another server because it is a contained database user. Its password is stored within the database, whereas **`reporting_user`** depends on the server-level **`reporting_login`**.

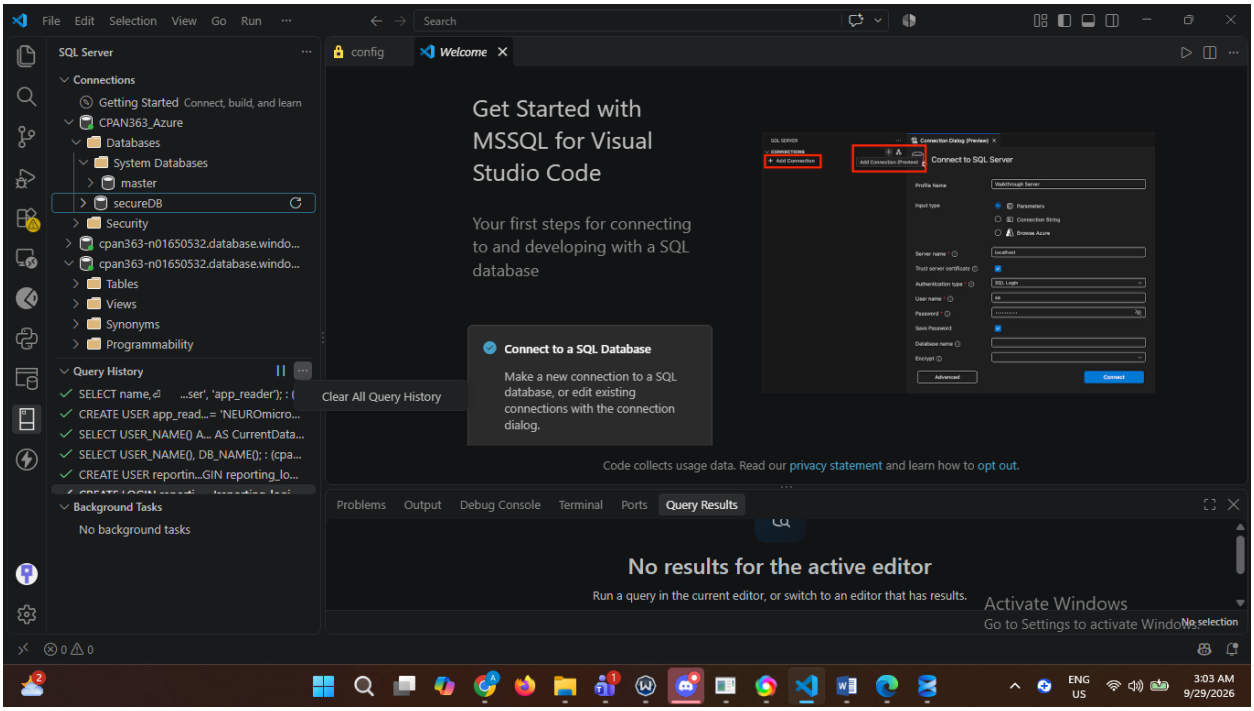
Microsoft Entra ID set up



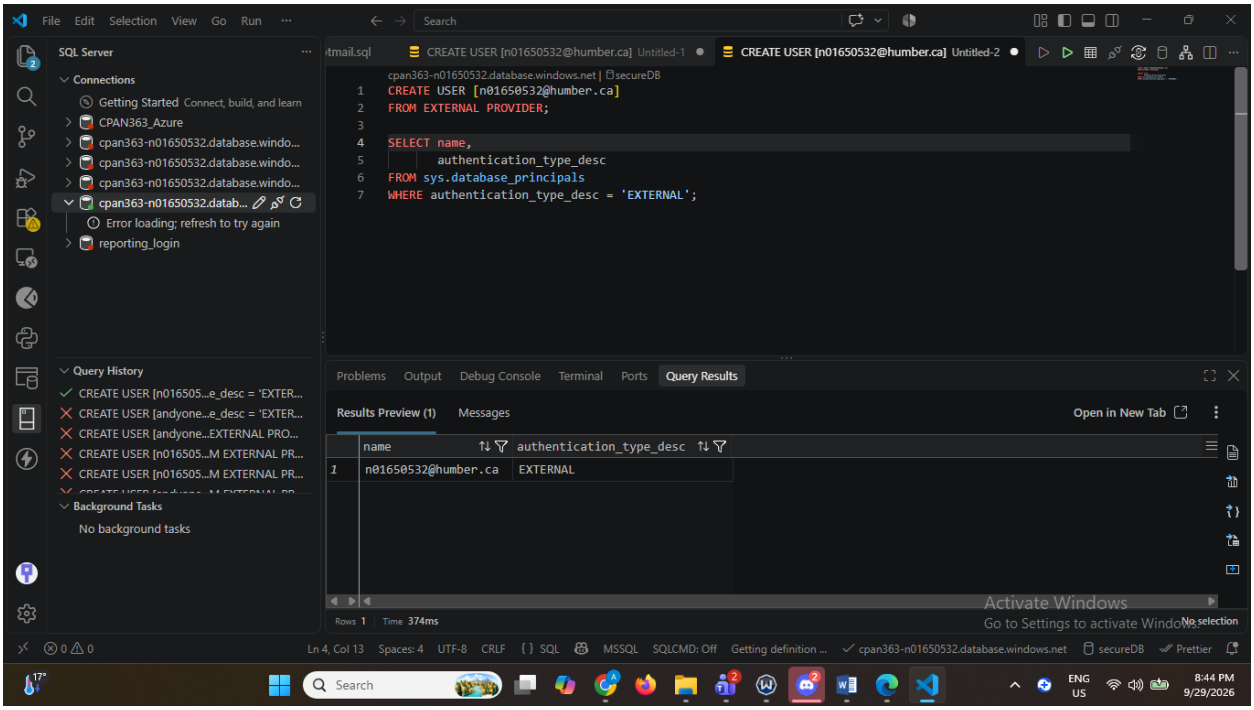



SQL Server
Connections
Getting Started Connect, build, and learn
CPAN363_Azure
Databases
System Databases
master
secureDB
Security
cpan363-n01650532.database.windo...
reporting_login
Query History
Background Tasks
No background tasks
Browse Azure
Browse Fabric
Server name *
cpan363-n01650532.database.windows.ne
Port
1433
Trust server certificate
Authentication type *
Microsoft Entra ID - Universal with MFA support
Visual Studio Code
The extension 'SQL Server (mssql)' wants to sign in using Microsoft.
Allow
Cancel
Encrypt
Mandatory
Advanced
Test connection
Connect
Import from Azure Data Studio
Recent Connections
cpan363-n01650532.database.windows.net, secureDB (andyonebox)
reporting_login
cpan363-n01650532.database.windows.net, master (andyonebox)
Saved Connections
CPAN363_Azure
cpan363-n01650532.database.windows.net, secureDB (andyonebox)
Problems
Output
Debug Console
Terminal
Ports
Query Results
No results for the active editor
Run a query in the current editor, or switch to an editor that has results.
Activate Windows
Go to Settings to activate Windows.
ENG US
2:48 AM
9/29/2026


HUMBER POLYTECHNIC
n01650532@humber.ca
Approve sign in request
Open your Authenticator app and approve the request. Enter the number if prompted.
16
Didn't receive a sign-in request? Swipe down to refresh the content in your app.
I can't use my Microsoft Authenticator app right now
For further assistance, contact Humber ITS
Terms of use
Privacy & cookies
Activate Windows
Go to Settings to activate Windows.
Cancel
ENG US
2:49 AM
9/29/2026

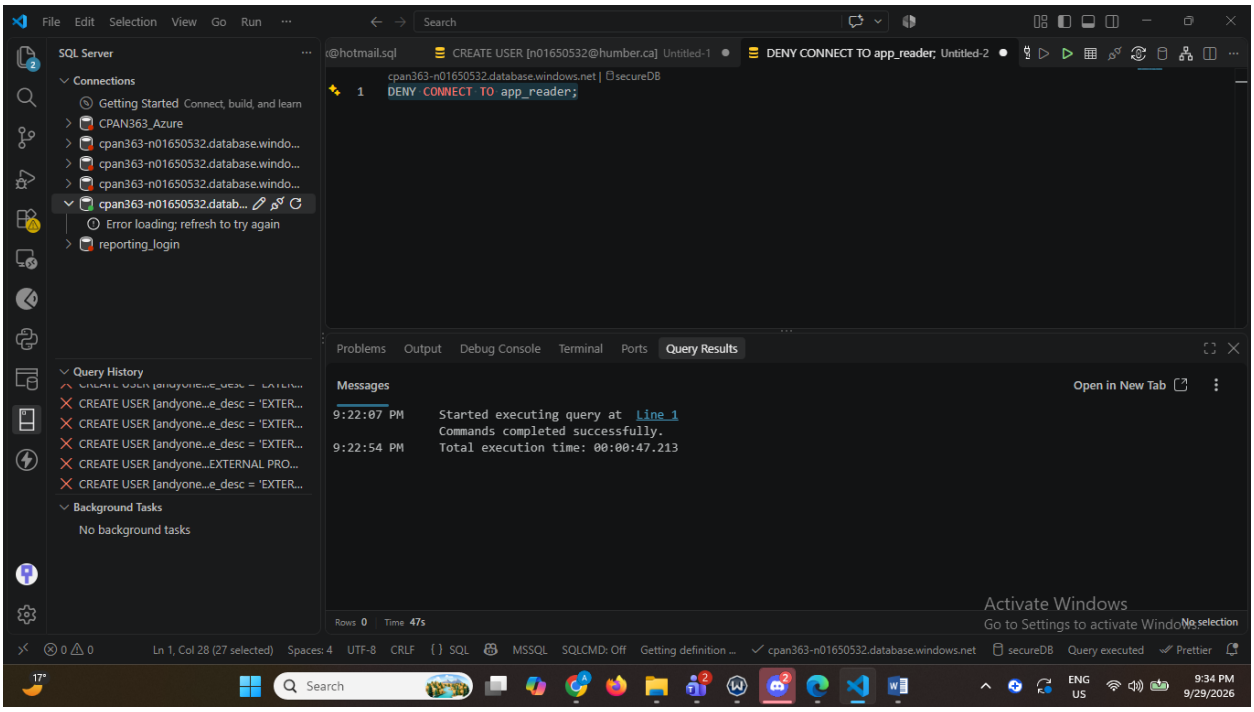

SQL Server
Connections
Getting Started Connect, build, and learn
CPAN363_Azure
Databases
System Databases
master
secureDB
Security
Tables
Views
Synonyms
Programmability
Query History
Clear All Query History
Background Tasks
No background tasks
Welcome
Get Started with MSSQL for Visual Studio Code
Your first steps for connecting to and developing with a SQL database
Connect to a SQL Database
Make a new connection to a SQL database, or edit existing connections with the connection dialog.
Connect to SQL Server
Code collects usage data. Read our privacy statement and learn how to opt out.
Problems
Output
Debug Console
Terminal
Ports
Query Results
No results for the active editor
Run a query in the current editor, or switch to an editor that has results.
Activate Windows
Go to Settings to activate Windows.
No selection
ENG
US
3:03 AM
9/29/2026

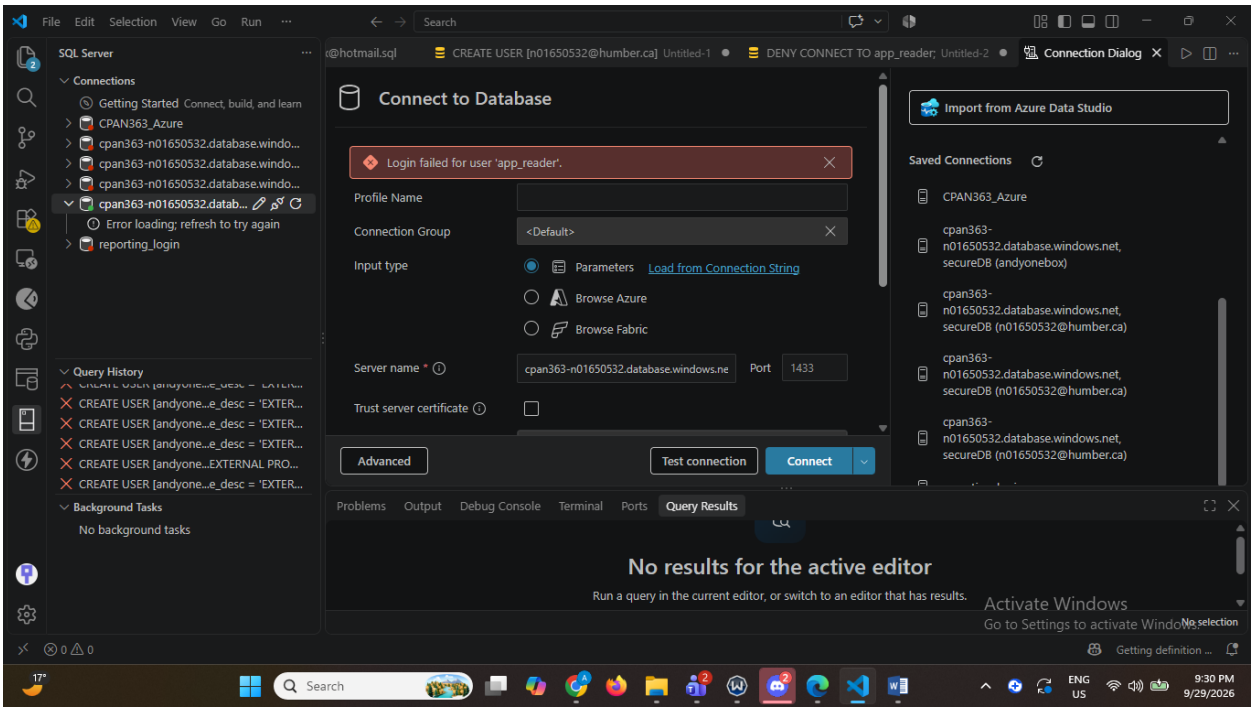
Part 3.3


SQL Server
Connections
Getting Started Connect, build, and learn
CPAN363_Azure
cpan363-n01650532.database.windo...
cpan363-n01650532.database.windo...
cpan363-n01650532.database.windo...
cpan363-n01650532.datab...
Error loading; refresh to try again
reporting_login
Query History
CREATE USER [n016505...e_desc = 'EXTER...
CREATE USER [andyone...e_desc = 'EXTER...
CREATE USER [andyone...EXTERNAL PRO...
CREATE USER [n016505...M EXTERNAL PR...
CREATE USER [n016505...M EXTERNAL PR...
Background Tasks
No background tasks
CREATE USER [n01650532@humber.ca] Untitled-1
CREATE USER [n01650532@humber.ca] Untitled-2
cpan363-n01650532.database.windows.net | secureDB
CREATE USER [n01650532@humber.ca]
FROM EXTERNAL PROVIDER;
SELECT name,
authentication_type_desc
FROM sys.database_principals
WHERE authentication_type_desc = 'EXTERNAL';
Problems
Output
Debug Console
Terminal
Ports
Query Results
Results Preview (1)
Messages
Open in New Tab
name
authentication_type_desc
n01650532@humber.ca
EXTERNAL
Rows 1
Time 374ms
Activate Windows
Go to Settings to activate Windows.
No selection
Ln 4, Col 13
Spaces: 4
UTF-8
CRLF
SQL
MSSQL
SQLCMD: Off
Getting definition ...
cpan363-n01650532.database.windows.net
secureDB
Prettier
ENG
US
8:44 PM
9/29/2026

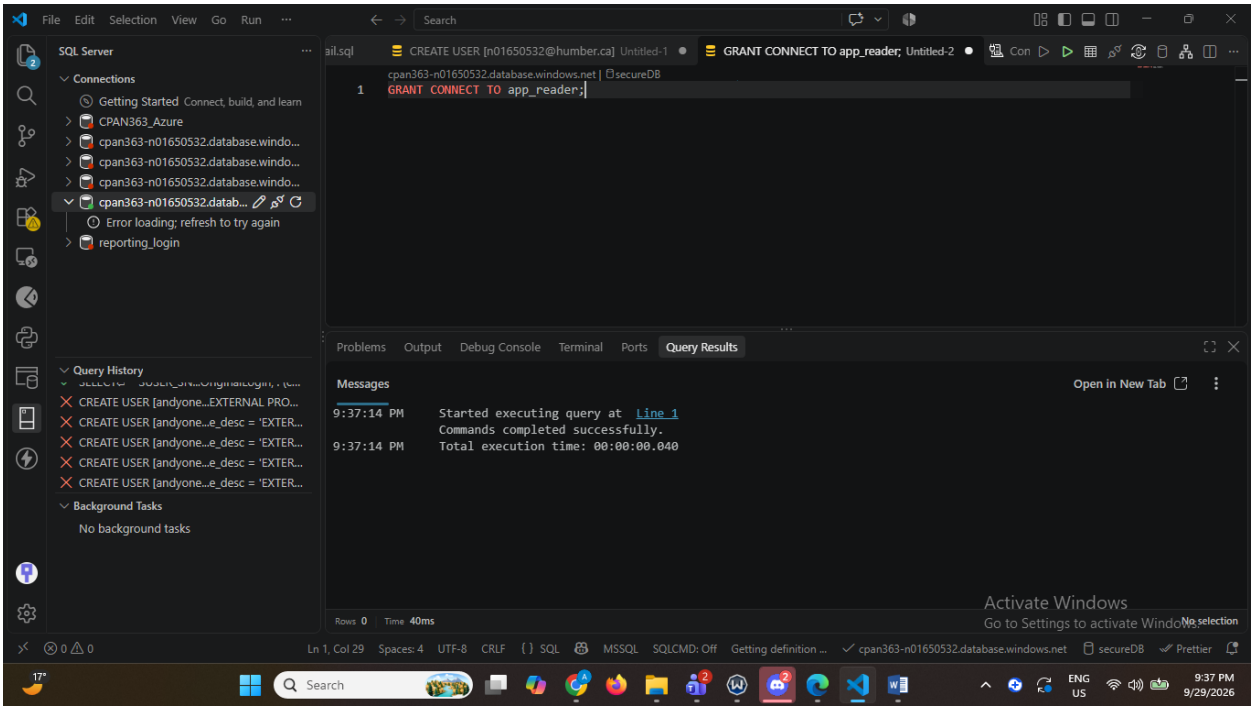
Part 4

4.1 Block app_reader from this database, then prove it worked.

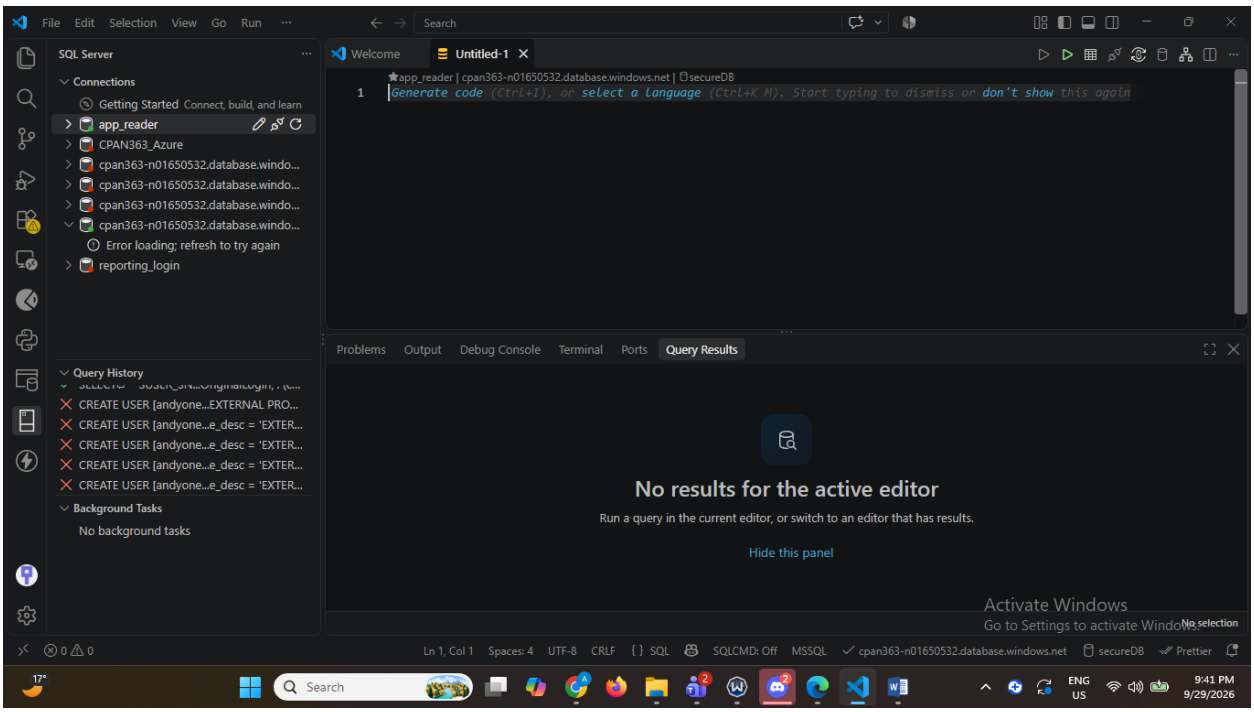




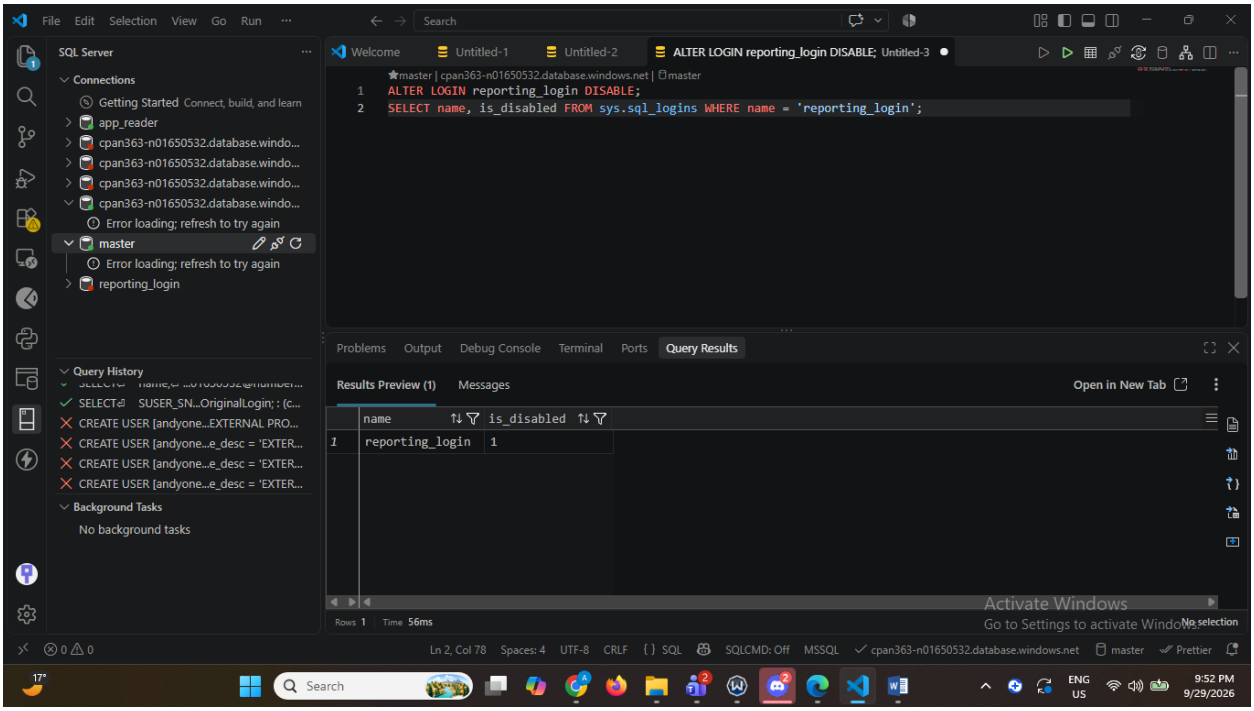
4.2 Let them back in and confirm.



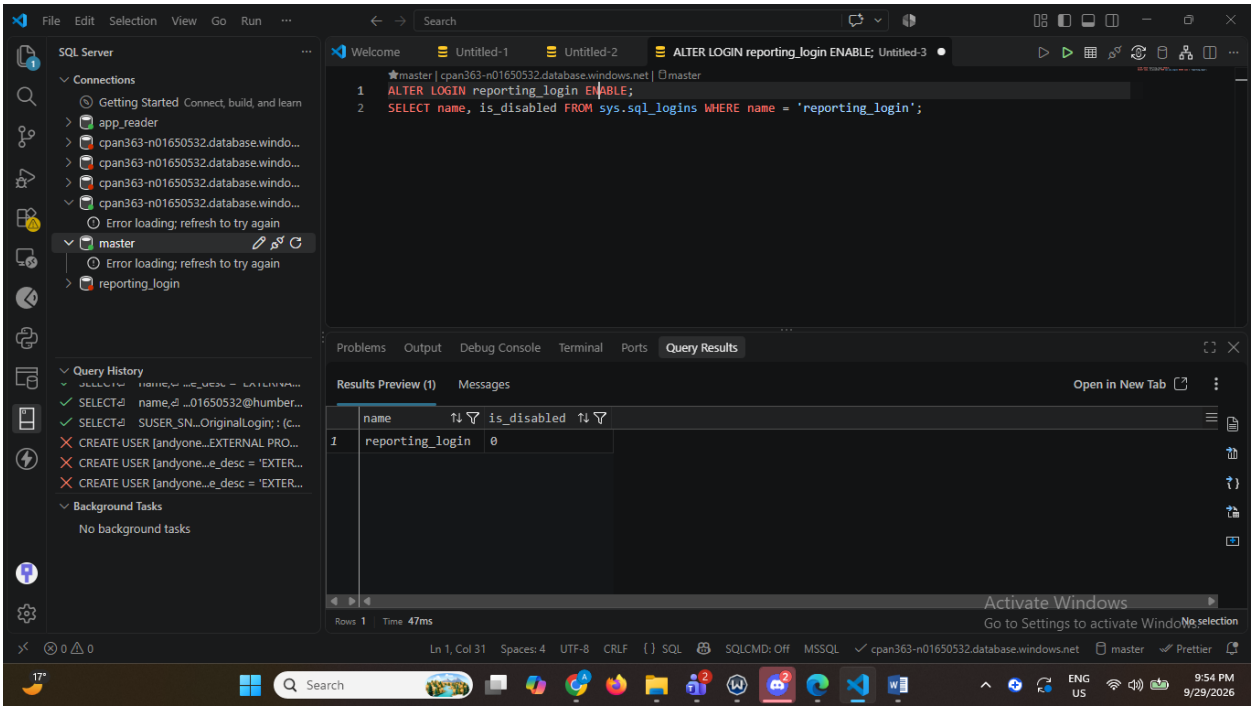
App_reader connection succeeded



4.3 Switch off the login at the server level,



then switch it back on.



4.4 Written explanation

`DENY CONNECT` prevents a user from connecting to a specific database, while **`ALTER LOGIN DISABLE`** disables the login at the server level. **`ALTER LOGIN DISABLE`** affects the login across databases because it disables the server-level login itself.

Part 5 — Think it through

Reception staff: I would use a Microsoft Entra account because reception staff are individual users who need controlled access to the database.

Doctors: I would use a Microsoft Entra account because each doctor is an individual user and their access should be associated with their own identity.

Website: I would use a contained database user because the website is an application that needs its own database credentials rather than a personal account.

5.2

The website should not use a personal database account. It should use an application/service identity with only the database permissions it requires.

5.3

When a doctor leaves, I would first disable or remove their database access. I would then verify that they can no longer connect, review their assigned permissions, remove unnecessary permissions, and confirm that access has been completely revoked.